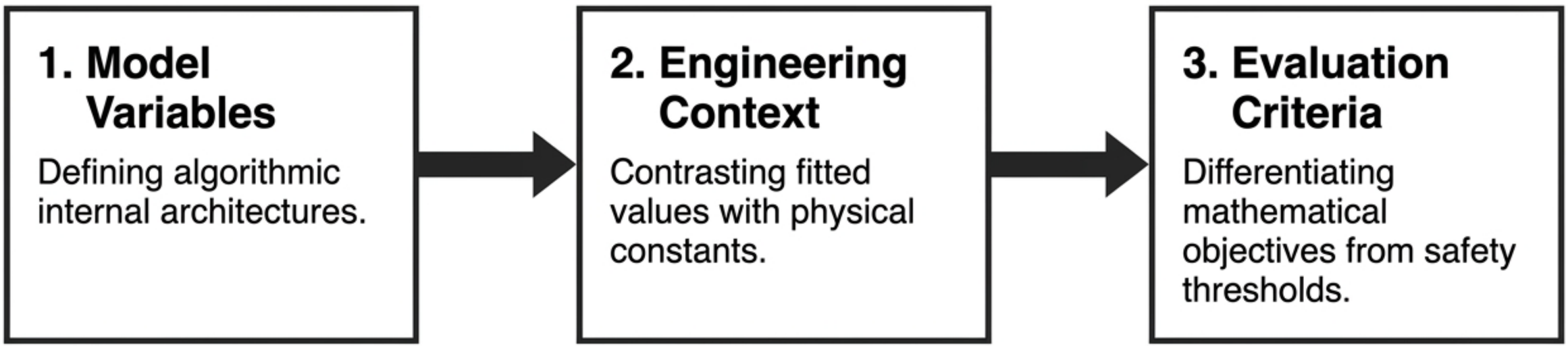
Module 6, Lesson 2

AI 101 for Mechanical Engineers: Parameters and hyperparameters

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Structural Architecture



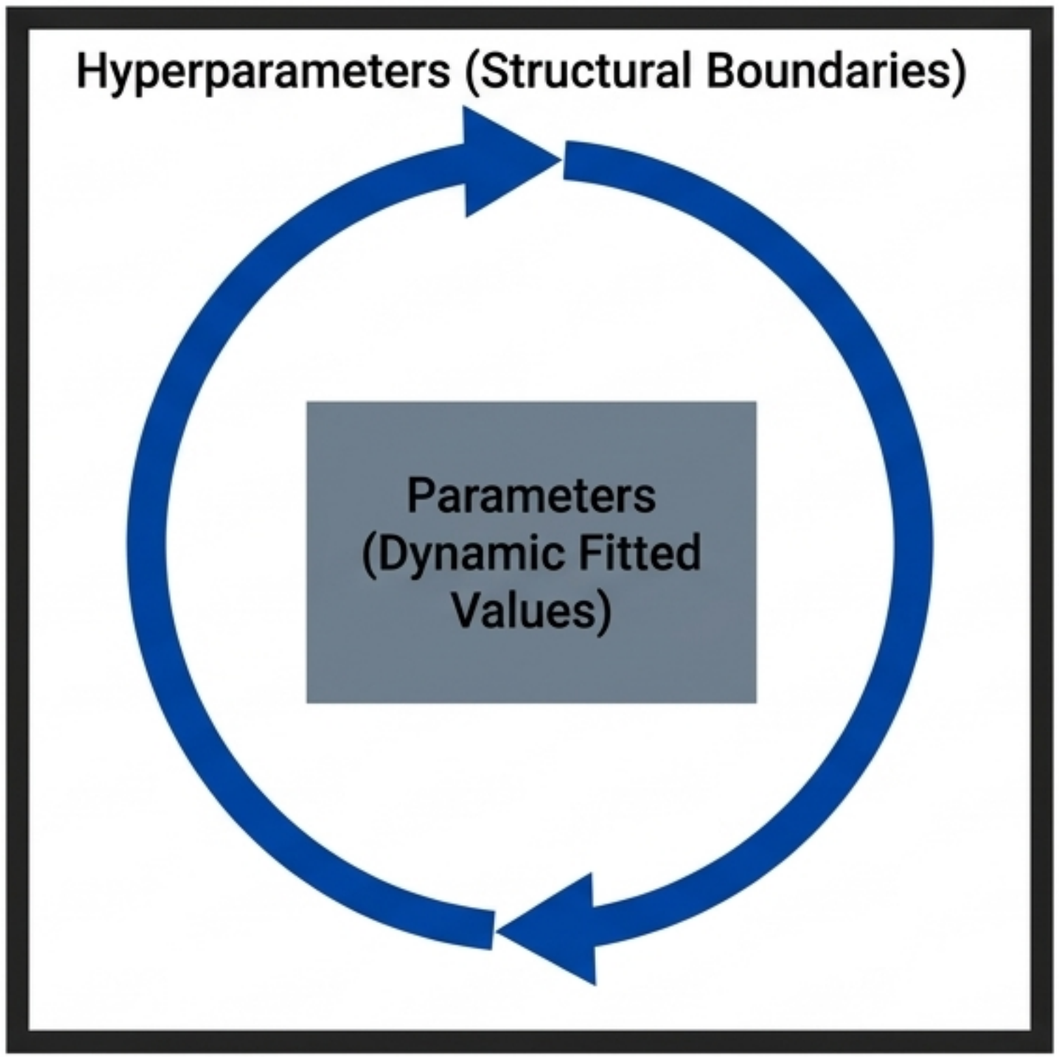
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1 Model Variables

Establish the distinction between internal architectural boundaries and algorithmically fitted values.

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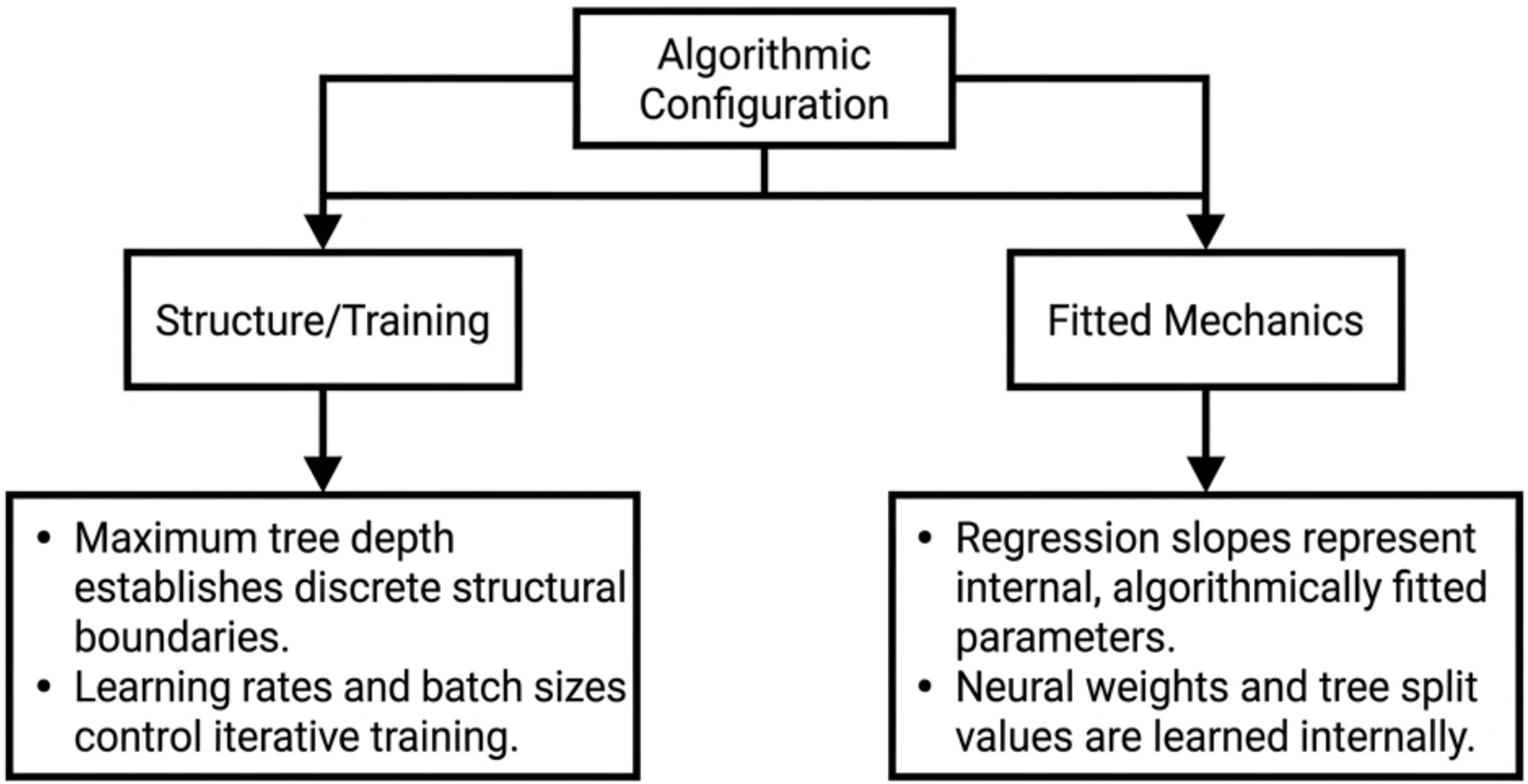
The Hierarchical Fitting Model



- Hyperparameters govern base model structure or training rules.
- Hyperparameters are usually selected through independent validation protocols.
- Parameters are specific numerical values fitted within models.
- Parameters are acquired through the internal learning operation.

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Variable Categorization Examples



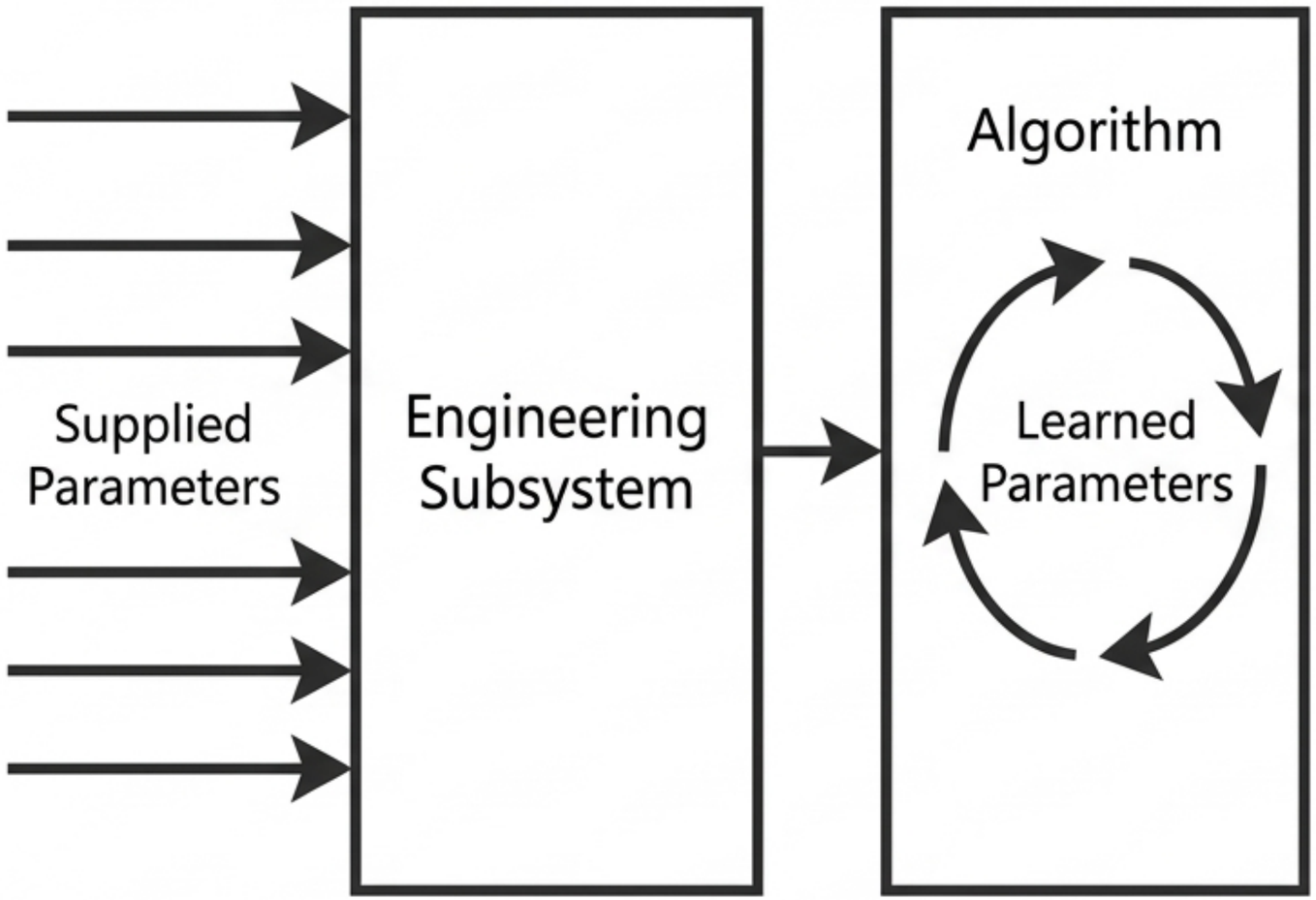
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2 Engineering Context

Contrast algorithmic learned variables with supplied physical engineering constants.

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Learned versus Supplied Variables



- Variable categorization depends strictly on the acquisition method.
- Physical efficiencies act as supplied engineering model assumptions.
- A supplied engineering parameter is **not** a machine learning parameter.
- Machine learning parameters require mathematical estimation during fitting.

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Diagnostic Comparison Matrix

Concept	Functional Role	Acquisition Method	Technical Example
Parameter	Fitted model value	Internal learning operation	Regression slope / intercept
Hyperparameter	Structure or training setting	External validation choice	Tree-depth limit / batch size
Supplied Parameter	Physical system assumption	Prior engineering specification	Fixed physical efficiency

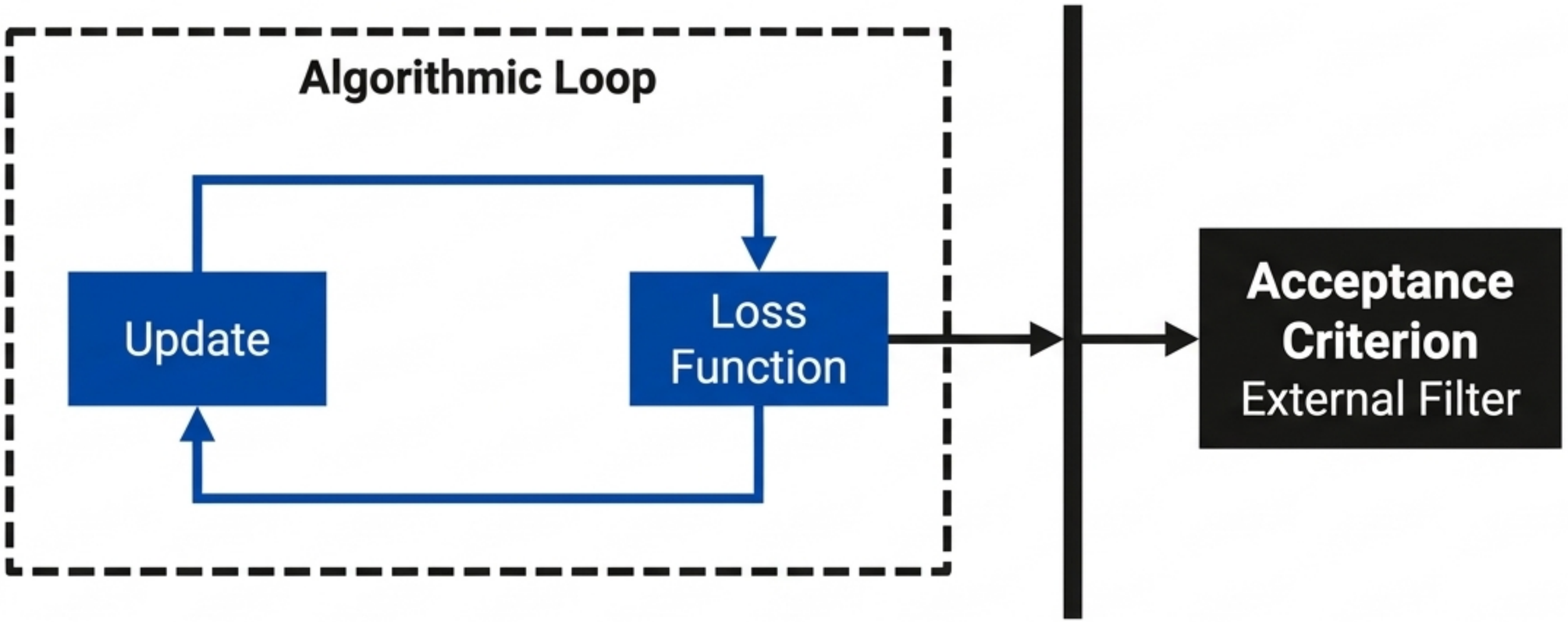
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3 Evaluation Criteria

Formalize the critical distinction between internal mathematical optimization and external engineering safety limits.

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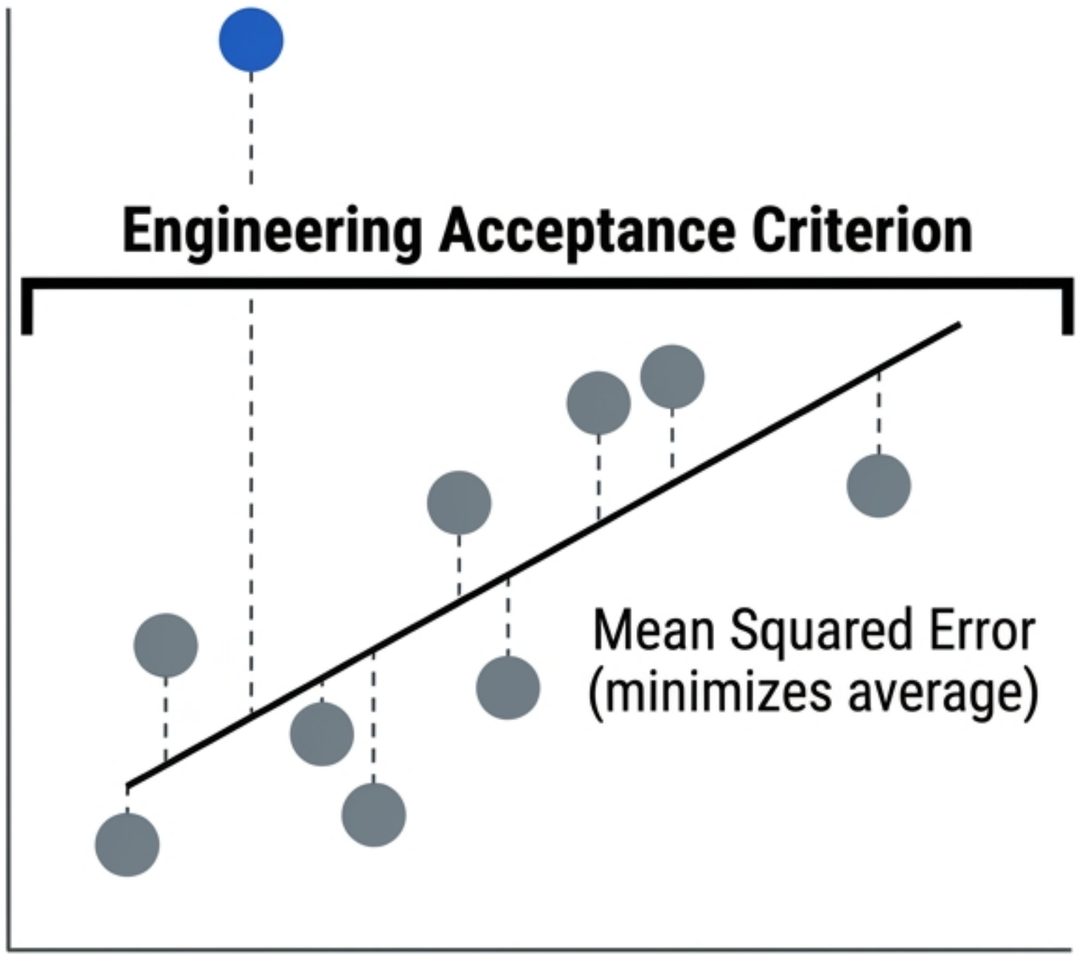
Internal Objectives vs. External Safety



- Loss functions quantify mathematical training disagreement or structural penalty.
- A training loss acts as an internal algorithmic objective.
- Acceptance criteria define external, non-negotiable engineering safety requirements.
- Mathematical loss and physical acceptance metrics frequently differ fundamentally.

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Loss versus Acceptance Constraints



- Mean squared error strictly favours aggregate average performance.
- Optimizing average aggregate metrics masks rare, catastrophic individual failures.
- Engineering acceptance demands absolute boundary constraint verification.
- Minimizing mathematical loss cannot guarantee operational system safety.

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Key Points

1. Hyperparameters establish structural boundaries; parameters represent dynamic, internally fitted values.

2. Algorithmic learned parameters differ fundamentally from fixed physical engineering constants.

3. Mathematical loss objectives do not inherently satisfy external engineering acceptance criteria.

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